

NLP for CounterSpeech against Hate and Misinformation



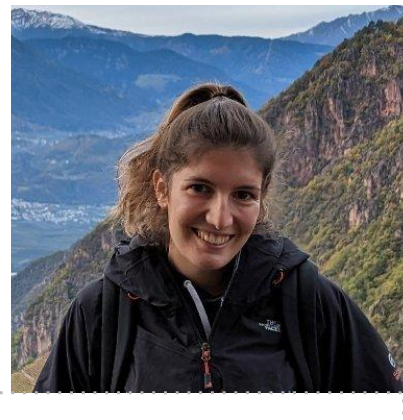
Organisers



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Program

09:30 11:00	CounterSpeech Tutorial Marco Guerini
11:00 11:15	Short BREAK
11:15 12:30	Hands On CounterSpeech Daniel Russo

Resources and Papers



Scan this to download the full list of references for today's presentations. The same list is also available in the tutorial website.

https://linktr.ee/CLiCit2026_Tutorial

Presentation Outline

01

What is CounterSpeech?

Introduction, fundamental definitions, and core communicative dynamics.

02

CS Against Hate Speech

Strategies and tactics for directly confronting and mitigating online hostility.

03

CS Against Misinformation

Fact-focused refutations and correcting false or misleading narratives.

04

CS Against Hate+Misinfo

Synthesized methods that address both harmful speech and false claims.

05

Open Challenges

Current limitations, research gaps, and future frontiers in the field.



Content warning

This tutorial will contain unobfuscated examples some viewers may find offensive.

Section 01

What is CounterSpeech?

Introduction, fundamental definitions, and core communicative dynamics.

Content

STEP 01.1

The Context

Online hate & misinfo:
standard moderation
approaches.

STEP 01.2

CounterSpeech

CS as a proactive,
non-censorial way to
react to online harms.

STEP 01.3

CS Impact

Analyzing
effectiveness and
social impact.

STEP 01.4

CS & NLP

Natural Language
Processing for CS
intervention.

The Context

Countering - Standard Approaches

In a world full of hate and misinformation how to react?

Standard policies are based on **identify-and-sanction** strategies.



Muslims are invading us, we should isolate them.

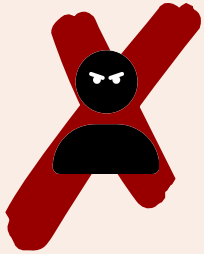
Content removal



Muslims are invading us we should isolate them.



Account suspension



Muslims are invading us, we should isolate them.

Shadow banning



Muslims are invading us, we should isolate them.

Countering - Contribution of AI

UP TO NOW: systems that are able to classify content (i.e. to tell if it is hateful or not)

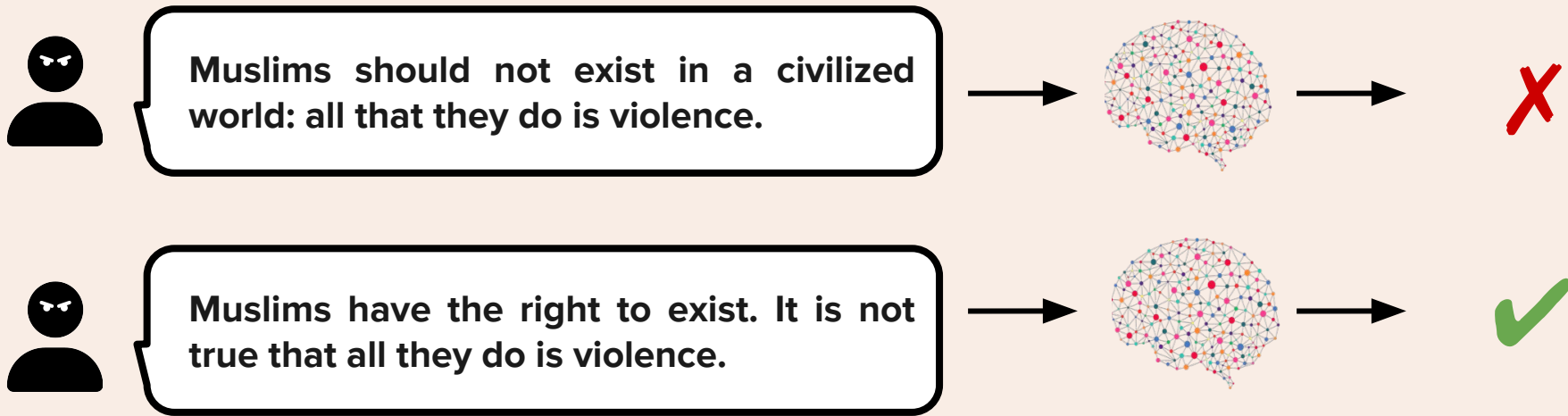
Countering - Contribution of AI

UP TO NOW: systems that are able to classify content (i.e. to tell if it is hateful or not)



Countering - Contribution of AI

UP TO NOW: systems that are able to classify content (i.e. to tell if it is hateful or not)



Countering - Contribution of AI

Simple word spotting is not enough most of the time. (Nozza, 2021)



Standard Approaches - Limitations

These approaches can be charged with censorship and overblocking. They can hinder Freedom of Speech.

They cannot be applied to **dangerous speech** - i.e. content that is stirring up hatred and divisiveness but does not fall into a formal definition of HS.

Dangerous Speech

True but spurious correlation...

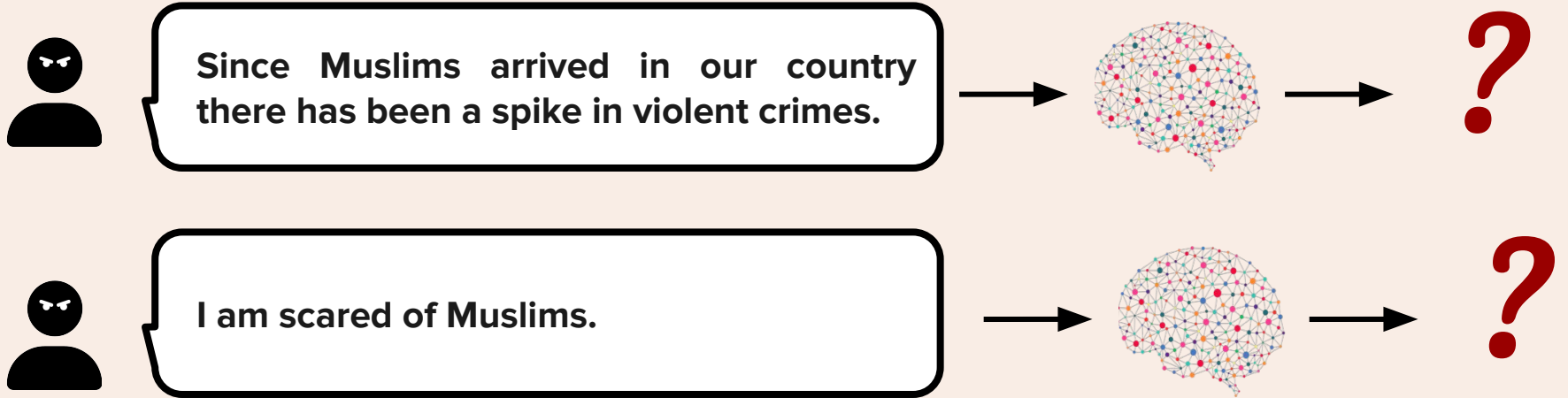
Dangerous Speech

True but spurious correlation...



Dangerous Speech

Expressing Emotions, ...



Counterspeech

Countering - CounterSpeech

One alternative strategy - receiving growing attention by the AI community - is to actually oppose hate content with CounterSpeech

CounterSpeech: **direct intervention in the discussion** to withstand hate messages using a non-aggressive textual response that offers feedback through fact-bound arguments. (Schieb and Preuss, 2016)

Important Note



CS uses a specialized language and it is written by **experts** following specific **guidelines**.

CounterSpeech Experts



NGO operators → against Hate



Fact checkers → against misinformation

Guidelines Example

4 HATE SPEECH

- Preserve the right to freedom of speech
- Counter stereotypes with credible evidence
- Alter the viewpoints of haters and bystanders
- Encourage mutual understanding
- Help de-escalating the conversation



Guidelines Example

4 HATE SPEECH

- Preserve the right to freedom of speech
- Counter stereotypes with credible evidence
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**PROSOCIAL
AIM**



CounterSpeech against Hate

CS can be very simple...



Women getting into the labour market has caused the downfall of Western civilisation, they should be at home raising children.

Stop spreading misogyny! We should all love each other.



CounterSpeech against Hate

CS can be very simple... but the most effective are carefully crafted.



Women getting into the labour market has caused the downfall of Western civilisation, they should be at home raising children.

I'd disagree, women should be able to choose what they do, and even if some women did want to stay at home, many don't have a choice anymore! Statistics show that it's almost impossible to support a family on one wage without going below the poverty line.



CounterSpeech against Hate

CS are also tailored to specific targets using specific anti-stereotypes arguments.



Jews control the banks, Hollywood, even the US government.

This claim is just a byproduct of “The Protocols of the Elders of Zion” a late 19th-century forgery about a supposed global Jewish conspiracy. Jews were accused of desecrating the communion bread and spreading the plague. Nowadays they are accused of controlling Hollywood and the banks.



CounterSpeech against Hate

CS are also tailored to specific targets using specific anti-stereotypes arguments.



Muslims are all violent. We let them in and this is how they pay us back!

It is wrong to generalize and discriminate based on religion, moreover the vast majority of Muslims worldwide condemn violence. Anti-Muslim groups frequently cherry-pick passages from the Quran to cite as evidence that Islam promotes violence.



CounterSpeech against Misinformation

Also known as *verdicts*: short text that evaluates the truthfulness of a claim and explains the reasoning behind that assessment.



11,000 of 13,000 knife attacks in London were carried out by Muslim migrants.

This claim is baseless as information on offenders' religion and nationality is not held by the authorities. Regardless, the claim is implausible.



CounterSpeech against Hate and Misinformation



Did you know that 11,000 of the 13,000 knife attacks in London were committed by Muslim immigrants? We let them in and this is how they pay us back!

This claim is baseless as information on offenders' religion and nationality is not held by the authorities. Moreover, it's not correct to make such a generalization about Muslims. Anti-Muslim groups frequently cherry-pick passages from the Quran to cite as evidence that Islam promotes violence.



CounterSpeech against Hate and Misinformation



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Misinformation component

This claim is baseless as information on offenders' religion and nationality is not held by the authorities. Moreover, it's not correct to make such a generalization about Muslims. Anti-Muslim groups frequently cherry-pick passages from the Quran to cite as evidence that Islam promotes violence. Yet the Quran includes many passages about tolerance, peace and love.



CounterSpeech against Hate and Misinformation



Did you know that 11,000 of the 13,000 knife attacks in London were committed by Muslim immigrants? We let them in and this is how they pay us back!

Hateful
component

This claim is baseless as information on offenders' religion and nationality is not held by the authorities. Moreover, it's not correct to make such a generalization about Muslims. Anti-Muslim groups frequently cherry-pick passages from the Quran to cite as evidence that Islam promotes violence. Yet the Quran includes many passages about tolerance, peace and love.



CounterSpeech Impact

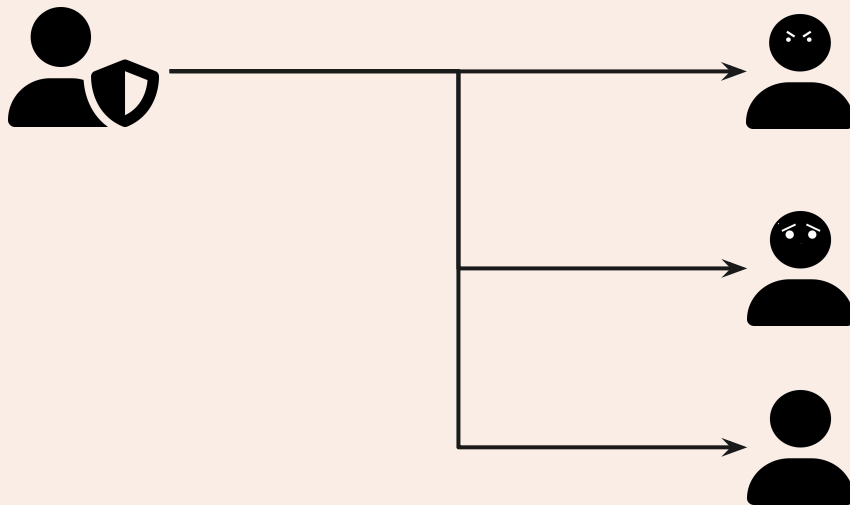
Hate Recipients

The target of the hate claim is the minority target, even if other interlocutors can be involved.



CounterSpeech Recipients

CounterSpeech has many different addressee.



Perpetrator

Directly challenging the speaker to stop or rebutting their claims.

Targets

Providing support, reducing emotional harm, and showing solidarity.

Bystanders

Influencing the silent majority and establishing healthy community norms.

Social Impact Measures

Evaluating how subjects interact with counterspeech online

- Participants willingness to intervene against hate (Leonhard et al., 2018)
- The amount of emerging alternative perspectives

Metrics

- Survey questions
- Bounce rate, exit rate, geo-location analysis
- Number of likes, views, and shares counterspeech posts receive

Behavioural Change Measures

Evaluating whether subjects change their observable behaviour towards victims before and after exposure to counterspeech

- Sentiment analysis (Hangartner et al., 2021)
- Levels of verbal aggression (Bilewicz et al., 2021)
- The proportion of hate speech in online discourse (Garland et al., 2020)
- The number of times users violate Facebook policies (Saltman et al., 2021)
- The likelihood of users continuing to use racial slurs following sanctions or penalties (Munger, 2017)

Attitude Change Measures

Evaluating whether subjects change their underlying attitudes or intentions

- The extent to which participants legitimized violence
- Support for extremist groups
- Intentions after counterspeech intervention

Metrics

- Psychological testing
- Survey, interviews
- Content analysis

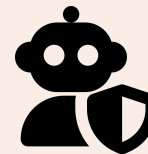
CounterSpeech & NLP

Why NLP for CounterSpeech

- NGOs and Fact Checker manually write CounterSpeech
- Time and resource-consuming
- NLG can partially automate this task with Language Models

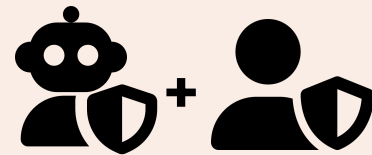
CounterSpeech Generation usage

Either complete automation (deprecated) ...



CounterSpeech Generation usage

Or human supervision/collaboration



CounterSpeech and Dialogue

At the beginning of the NLP research on CS the focus has been on a single response...



CounterSpeech and Dialogue

... but in real interventions experts often engage in a fully fledged dialogue.



Section 02

CS Against Hate Speech

Strategies and tactics for directly confronting and mitigating online hostility.

Content

STEP 02.1

Design your task

Classification, Selection
or Generation?

STEP 02.2

Select the data

Collect it or choose
from existing datasets?

STEP 02.3

CS evaluation

Automatic or Human
Evaluation?

Design your task

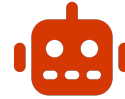
Main tasks



CLASSIFICATION



SELECTION



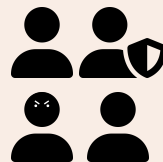
GENERATION

Classification



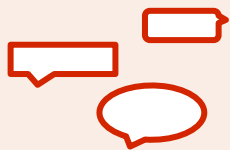
CounterSpeech detection

- > Data collection
- > Understand CounterSpeech dynamics



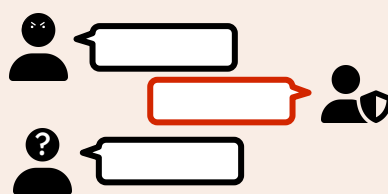
User classification

- > Allow platforms' early intervention (Mathew et al., 2020)



Strategy classification

- > Analyse effectiveness
- > Develop fine-grained responses



Hater's reentry

- > Analysis of hater's behavior to understand CounterSpeech effectiveness (Song et al., 2025)

Classification

Why does CounterSpeech classification usually fail?

- Complex text
(Goffredo et al., 2022; Albanyan and Blanco, 2022; Yu et al., 2022)
- Lexical overfitting to specific terms
(Vidgen et al., 2020; Yu et al., 2022)
- Annotation errors
(Vidgen et al., 2020)

Classification

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- Lexical overfitting to specific terms
(Vidgen et al., 2020; Yu et al., 2022)
- Annotation errors
(Vidgen et al., 2020)



How to reduce errors?

- Including the context
(Vidgen et al., 2021; Yu et al., 2022; Albanyan and Blanco, 2022)
- Pretraining on similar tasks:
stance or emotion detection
(Yu et al., 2022; Albanyan and Blanco, 2022)
- Large dataset with high quality
annotation

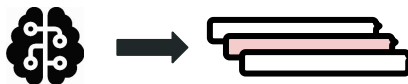
Selection

From existing dataset



Zhu and Bhat (2021)

Through over-generation



Chung et al. (2021c);
Akazawa et al. (2023)

From online content



Möhle et al. (2023);
Albanyan et al. (2023a)

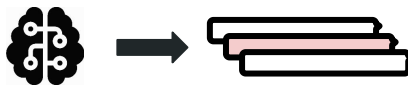
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Albanyan et al. (2023a)

🔧 Selecting the most appropriate response over a pool of gold or silver available CounterSpeech is preferable to filtering out non-CounterSpeech instances.

Generation

(some) desirable aspects



INFORMATIVENESS

Extracting knowledge
from external sources
→ generate



Correcting stereotypes alone
is not sufficient and can lead to
higher levels of violence (Carthy
and Sarma, 2023).

Generation

(some) desirable aspects



INFORMATIVENESS

Extracting knowledge
from external sources
→ generate



PERSONALITY AND STYLE

Persona profile,
argumentative
information, strategies



Correcting stereotypes alone
is not sufficient and can lead to
higher levels of violence (Carthy
and Sarma, 2023).



Empathy-based CounterSpeech
(Hangartner et al., 2021) or using a
particular strategy according to the
targeted minority (Mathew et al., 2019).

Generation

Relevant generation techniques

Fine-tuning

 It is the most employed approach. It allows for more adherent generations to the training data.

Prompting

 Prompting is computationally low-intensive, but clear and specific instructions are necessary.

Translation

 In low-resourced languages: translate from a strong model in another language (e.g. EN).

Select the data

DATASETS

Data Collection Strategies

CRAWLING

Scraping of social media

CROWDSOURCING

Non expert annotators

NICHESOURCING

Expert based data collection



SYNTHETIC DATA

Data fully generated by LLMs

HYBRID APPROACHES

Human and LLMs collaboration

Collect your own data

Strategy	Quantity	Conformity to guidelines	Diversity	Non-eph.
Crawling	✓	-	✓	-

on my way to f**k your b**ch.

But my b**ch is your mamma

Collect your own data

Strategy	Quantity	Conformity to guidelines	Diversity	Non-eph.
Crowdsourcing	✓	✓	-	✓

Like... I literally addressed this point and you completely ignore it, like a stupid retard n****r.

Stop using racist and medical conditions as an insult.

Collect your own data

Strategy	Quantity	Conformity to guidelines	Diversity	Non-eph.
Crowdsourcing	✓	✓	-	✓
Synthetic data	✓	✓	-	✓

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Crowdsourcing	✓	✓	-	✓
Nichesourcing	-	✓	✓	✓
Synthetic data	✓	✓	-	✓

SELECT THE DATA

Collect your own data

Strategy	Quantity	Conformity to guidelines	Diversity	Non-eph.
Hybrid approaches	✓			✓



Machine generation

Collect your own data

Strategy	Quantity	Conformity to guidelines	Diversity	Non-eph.
Hybrid approaches	✓	✓	✓	✓



Machine generation

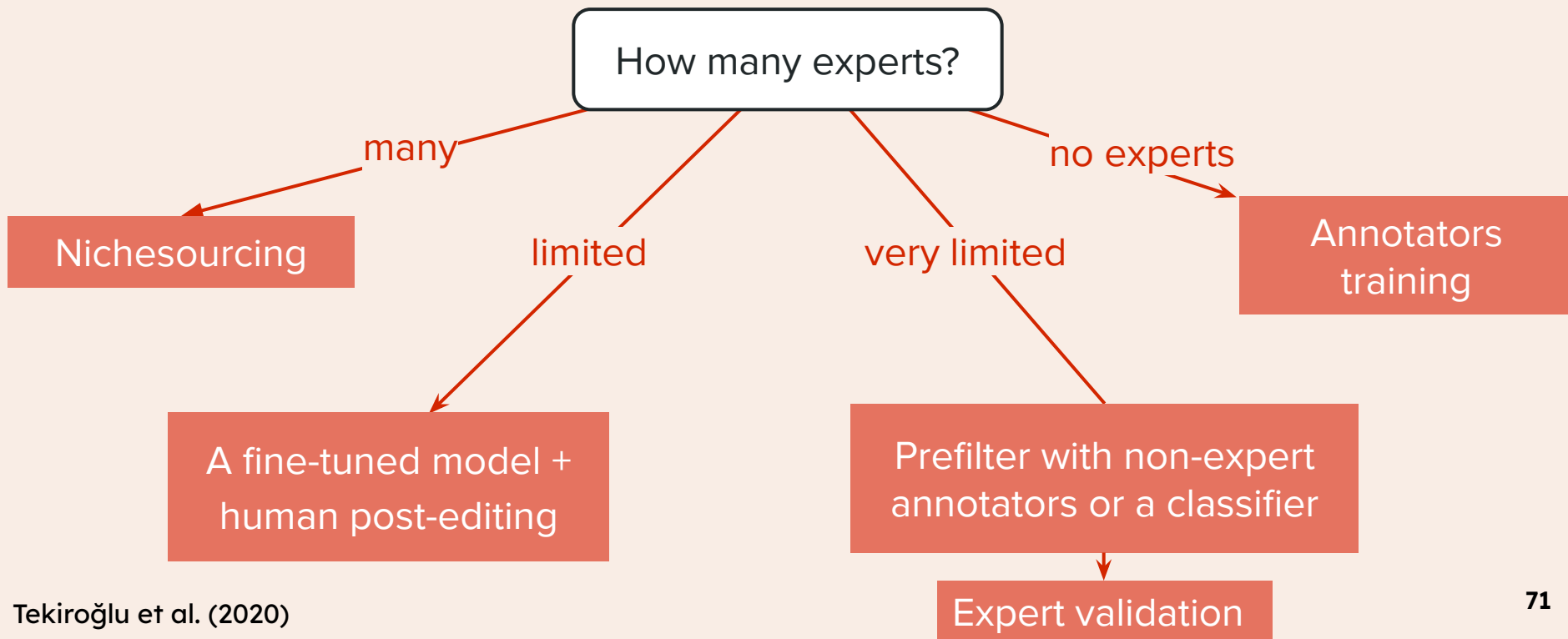


Human revision

Collect your own data

Strategy	Quantity	Conformity to guidelines	Diversity	Non-eph.
Crawling	✓	-	✓	-
Crowdsourcing	✓	✓	-	✓
Nichesourcing	-	✓	✓	✓
Synthetic data	✓	✓	-	✓
Hybrid approaches	✓	✓	✓	✓

Collect your own data



Choose from existing datasets

Quantity/quality trade-off

Dataset	Size	# CS	Interact.	Coll.	Source	Lang.	Tar.	Add.
Mathew et al. (2019)	13,924	6,898	Pairs + c.	Crawl.	YouTube	EN	✓	✓
Chung et al. (2019)	14,988	14,988	Pairs	Nich.	NGOs op.	EN/FR/IT	✓	✓
Qian et al. (2019)	16,845	29,388	Pairs + c.	Crowd.	Reddit, Gab	EN	-	-
Mathew et al. (2020)	1,290	1,290	Pairs	Crawl.	Twitter	EN	-	✓
Vidgen et al. (2020)	20,000	116	Single c.	Crawl.	Twitter	EN	✓	-
He et al. (2021)	2,290	517	Single c.	Crawl.	Twitter	EN	✓	✓
Vidgen et al. (2021)	27,494	220	Single c.	Crawl.	Reddit	EN	-	-
Chung et al. (2021b)	195	195	Pairs	Niches.	NGO op.	EN	✓	✓

Expert-based but small or covering only one target

Choose from existing datasets

Quantity/quality trade-off

Dataset	Size	# CS	Interact.	Coll.	Source	Lang.	Tar.	Add.
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Chung et al. (2021b)	195	195	Pairs	Niches.	NGO op.	EN	✓	✓

Big but no experts involved (lower quality)

Choose from existing datasets

Quantity/quality trade-off

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Chung et al. (2021b)	195	195	Pairs	Niches.	NGO op.	EN	✓	✓
Fanton et al. (2021)	5,003	5,003	Pairs	Hybr.	NGOs op.	EN	✓	-
Bonaldi et al. (2022a)	3,059	8,311	Dialog.	Hybr.	NGOs op.	EN	✓	-

Choose from existing datasets

4 Low-resourced languages →

- **European languages:** Italian (Goffredo et al., 2022), French (Chung et al., 2019) and Spanish (Furman et al., 2022 and 2023, Vallecillo-Rodríguez et al., 2023)

Choose from existing datasets

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- **European languages:** Italian (Goffredo et al., 2022), French (Chung et al., 2019) and Spanish (Furman et al., 2022 and 2023, Vallecillo-Rodríguez et al., 2023)
- **Non European languages:** Bengali and Hindi (Das et al., 2024), Modern Standard Mandarin (Bennie et al., 2025)

Choose from existing datasets

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- **Non European languages:** Bengali and Hindi (Das et al., 2024), Modern Standard Mandarin (Bennie et al., 2025)
- **X-low-resourced in Europe:** Basque (Bengoetxea et al., 2024)

Evaluation

Automatic evaluation



CLASSIFICATION

F1, precision, recall, accuracy,
confusion matrices



GENERATION

BLEU, ROUGE, Toxicity,
Repetition Rate, ...

Automatic evaluation



CLASSIFICATION

F1, precision, recall, accuracy, confusion matrices



GENERATION

BLEU, ROUGE, Toxicity, Repetition Rate, ...



Can capture the performance of a system at scale. Some lack interpretability and correlation with human evaluation (Belz and Reiter, 2006; Novikova et al., 2017).

Human evaluation



CLASSIFICATION

Judgements compared to classifiers, qualitative error analysis



GENERATION

Suitableness, informativeness, coherence, grammaticality

Human evaluation



CLASSIFICATION

Judgements compared to classifiers, qualitative error analysis



GENERATION

Suitableness, informativeness, coherence, grammaticality



CounterSpeech is subjective: enlist a diverse set of annotators (Waseem et al., 2017; Sap et al., 2019; Abercrombie et al., 2023) or users identifying as potential recipients of CounterSpeech is preferable. Problem: costly.

LLM as a judge?

- Jones et al. 2024: multi-aspect framework
- Zubiaga et al. 2024: JudgeLM, pairwise comparisons of CounterSpeech to assess overall quality

LLM as a judge?

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Both achieve good correlation with human judgement but no experts involved in the evaluation. It is a subjective task and agreement is low even between humans. Lack of a large-scale study.

Section 03

CS Against Misinformation

Fact-focused refutations and correcting false or misleading narratives.

Content

STEP 03.1

Background

Automated
Fact-checking and
Social correction

STEP 03.2

Datasets

Available Datasets for
CS against misinfo

STEP 03.3

CS Generation

Earliest and current
RAG approaches

STEP 03.3

CS evaluation

Automatic or Human
Evaluation?

Introduction

Definitions

MISINFORMATION

Intentional or **unintentional** promulgation of false or inaccurate information.

FACT-CHECKING

Process that aims at verifying that all the facts mentioned in a text or speech are correct.



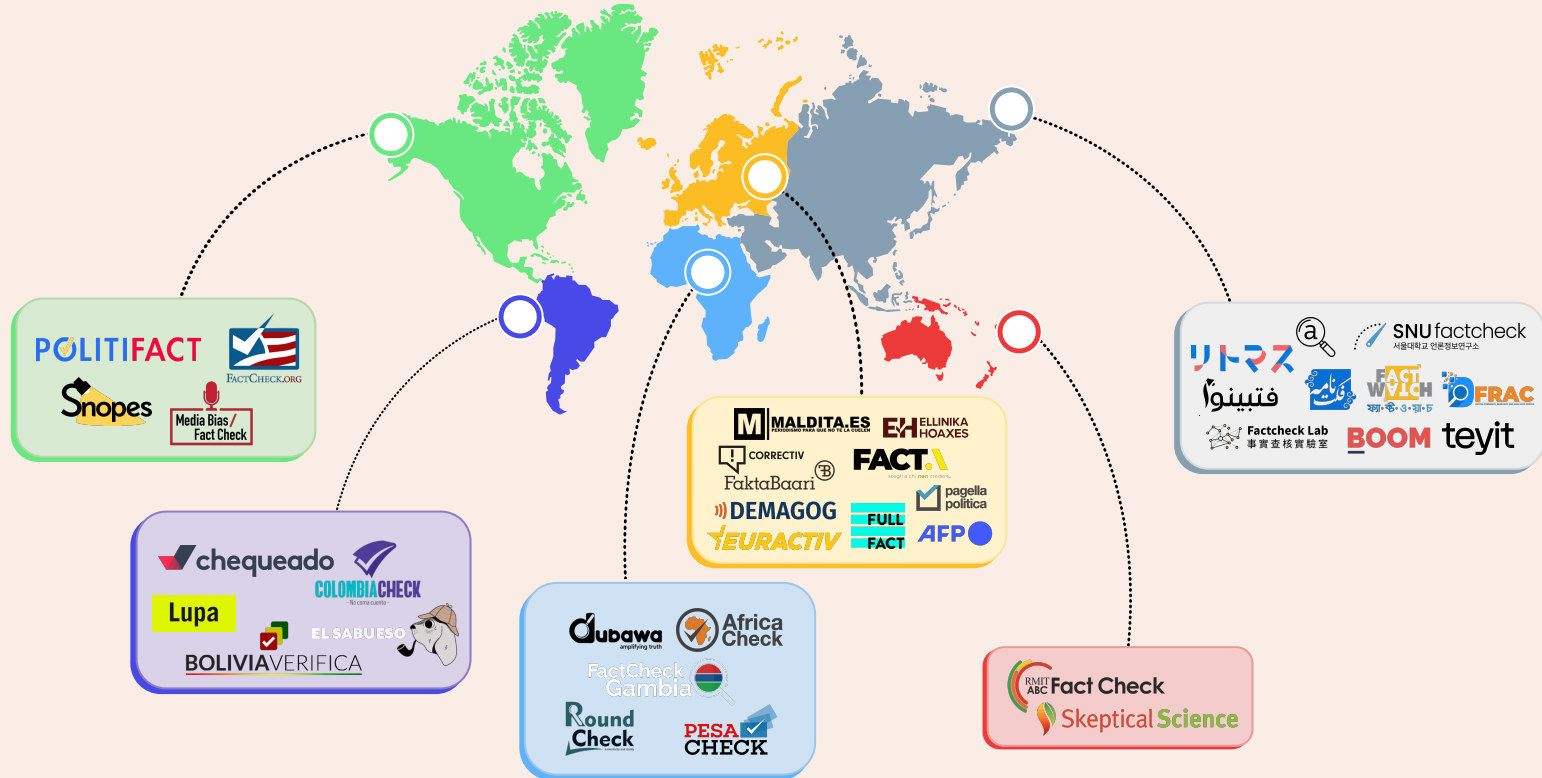
DISINFORMATION

Intentional promulgation of false or inaccurate information (e.g., fake news)

SOCIAL CORRECTION

Correction of misinformation conducted directly on social media by other social media users.

Fact-Checking Organisations



Automated Fact-Checking

Natural Language Processing has been proposed as a viable solution for partially automating fact-checking processes (Guo et al., 2022)

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CLAIM DETECTION



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Binary Classification or
importance ranking tasks

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Binary Classification or
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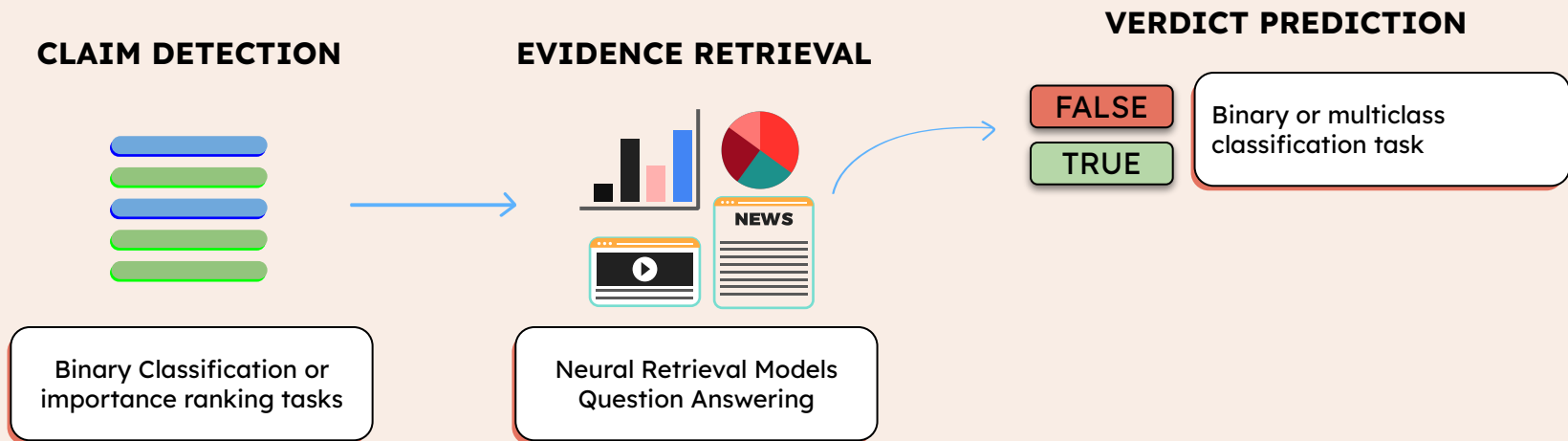
EVIDENCE RETRIEVAL



Neural Retrieval Models
Question Answering

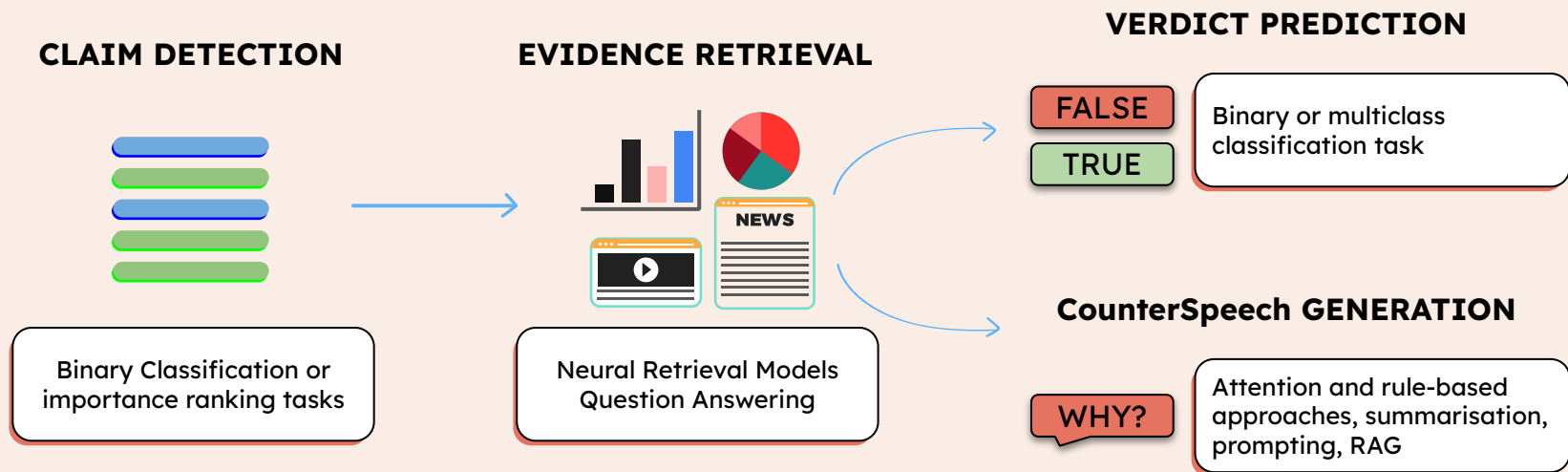
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Why CounterSpeech Generation?

1

Fasten the time consuming work of professional fact-checkers (Guo et al., 2022)

3

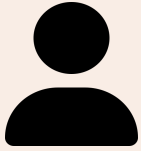
Common users do not engage with long fact-checking articles (Glensky et al., 2020)



2

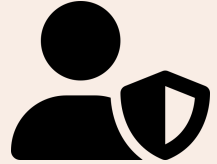
Simple classification might backfire and reinforce belief in false information (Lewandowsky et al., 2012)

Fact-Checking



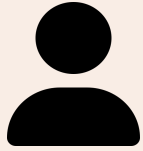
The vaccine manufacturers do not have liability.

Covid-19 vaccine manufacturers are immune to some, but not all, civil liability.



An effective CounterSpeech clearly explains why a claim is false, is easy to understand, and contains few arguments (Lewandowsky et al., 2012; Lombrozo, 2007; Sanna and Schwarz, 2006).

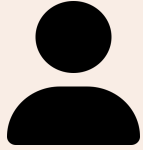
Fact-Checking



As someone who has lost a loved one due to vaccine complications, it makes my blood boil to think that the vaccine manufacturers have zero liability. How is this fair? They should be held accountable for any harm caused by their products. #vaccinesafety #justiceforvictims

Emotional state
Personal
Informal

Fact-Checking

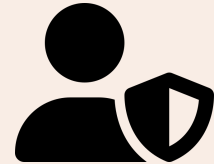


As someone who has lost a loved one due to vaccine complications, it makes my blood boil to think that the vaccine manufacturers have zero liability. How is this fair? They should be held accountable for any harm caused by their products. #vaccinesafety #justiceforvictims

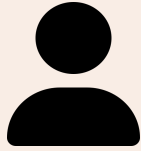
Emotional state
Personal
Informal

Impersonal
Formal
Apathetic

Covid-19 vaccine manufacturers are immune to some, but not all, civil liability.



Social Correction



As someone who has lost a loved one due to vaccine complications, it makes my blood boil to think that the vaccine manufacturers have zero liability. How is this fair? They should be held accountable for any harm caused by their products. #vaccinesafety #justiceforvictims

Emotional state
Personal
Informal

Empathetic
Informal
Polite

I can't even begin to imagine the pain your loss has caused you. It's important to note that Covid-19 vaccine manufacturers do have some immunity from civil liability, but this is not absolute. Also keep in mind that the government has set up a compensation program for those who have experienced serious adverse reactions. I hope this information helps, and I hope you do better soon. #vaccinesafety #compassionforall



Social Correction

The work of professional fact-checkers is crucial for countering misinformation (Guo et al., 2022)

It has been shown that **most debunking on Social media (96%) is carried out by ordinary users** through direct replies to misleading message (Wintersieck, 2017)

In the literature, this phenomenon is called **Social Correction** (Ma et al., 2023)

An effective CounterSpeech is characterised as being polite, empathetic, accessible, but also based on arguments and evidence.

Datasets

Data Collection Strategies

NICHESOURCING

Expert based data collection

SYNTHETIC DATA

Data fully generated by LLMs



CRAWLING

Scraping of Fact-checking websites

HYBRID APPROACHES

Expert and LLMs collaboration

Datasets

DATA COLLECTION STRATEGIES

	MANUAL	CRAWLING	SYNTHETIC	HYBRID	MULTI-LINGUAL	GENERAL DOMAIN	SOCIAL CORRECTION	MULTI-MODAL
LiarPlus (Alhindi et al., 2018)	✗	✓	✗	✗	✗	✓	✗	✗
e-FEVER (Stammbach & Ash, 2020)	✗	✗	✓	✗	✗	✓	✗	✗
PUBHEALTH (Kotonya & Tony, 2020)	✗	✓	✗	✗	✗	✗	✗	✗
FullFact (Russo et al., 2023b)	✗	✓	✗	✗	✗	✓	✗	✗
Liar++ (Russo et al., 2023b)	✗	✓	✗	✗	✗	✓	✗	✗
MOCHEG (Yao et al., 2023)	✗	✓	✗	✗	✗	✓	✗	✓
VerMouth (Russo et al., 2023a)	✗	✗	✗	✓	✗	✓	✓	✗
ExClaim (Zeng & Gao, 2024)	✗	✓	✗	✗	✗	✓	✗	✗
MisinfoCorrect <i>in-the-wild</i> (He et al., 2024)	✗	✓	✗	✗	✗	✗	✓	✗
MisinfoCorrect <i>crowdsourced</i> (He et al., 2024)	✗	✗	✗	✓	✗	✗	✓	✗
RU22Fact (Zeng et al., 2024)	✗	✓	✗	✗	✓	✗	✗	✗
EuroVerdict (Russo et al., 2025)	✓	✗	✗	✗	✓	✓	✗	✗

CS Generation Approaches

Earliest Approaches

Systems that incorporate components that human experts can interpret as justifications.

ATTENTION-BASED

Select evidence tokens with higher attention weights as explanations.

(Popat et al., 2018; Shu et al., 2019; Yang et al., 2019; Wu et al., 2020; Lu and Li, 2020)

RULE-BASED

Construct decision-making processes fully interpretable by humans, using Horn rules to mine explanations from knowledge bases.

(Gad-Elrab et al., 2019; Ahmadi et al., 2019)

Earliest Approaches

ATTENTION-BASED

[False] : Coca-Cola's original diet cola drink, TaB, took its name from an acronym for "totally artificial beverage."

Article Source: foxnews.com

the first diet colas being the first in 1952 cocacola execs at that time were hesitant to affix the term diet to cocacola so the name tab was chosen as a tribute to those who were keeping tab of their weight according to cola legend the drink was actually dubbed tab as an acronym for totally artificial beverage a great story which unfortunately cocacola says is completely untrue the name was actually chosen by computer and market research the saccharin scandal in the 70s did its damage and the introduction of diet coke in the early 1980s pushed tab even

(Popat et al., 2018)

RULE-BASED

```
FALSE : almaMater(Michael White, UT Austin)
← employer(Michael White, UT Austin)
← occupation(Michael White, UT Austin)
← almaMater(Michael White, Abilene
  Christian Univ.), almaMater(Michael
  White, Yale Divinity School)
```

(Ahmadi et al., 2019)

LIMITATIONS

Both approaches lack in providing human accessible CounterSpeech (**readability**).

Attention-based approaches are not fully able to capture model reasoning (**faithfulness**)

Current Approaches

Leverage (L)LM capabilities to improve readability and factuality of the CounterSpeech

SUMMARIZATION

CounterSpeech is obtained through the summarization of fact-checking articles.

Extractive (Atanasova et al., 2020), **abstractive** (Kotonya and Toni, 2020; Zeng et al., 2024), or **hybrid** summarization approaches (Russo et al., 2023a,b) were proposed.

LIMITATIONS

Extractive summaries might produce grammatical inconsistencies.

Abstractive approaches suffer model hallucination.

Summarization approaches relies on the existence of fact-checking articles.

Current Approaches

Leverage (L)LM capabilities to improve readability and factuality of the CounterSpeech

PROMPTING

Leverage in-context-learning abilities of pretrained LLMs to directly provide a CounterSpeech of a given claim given the information provided in input.

Useful both in monolingual and multilingual settings (e.g., Stammbach & Ash, 2020; Russo et al., 2025)

LIMITATIONS

LLMs are sensitive to prompt variations and might produce factual inaccuracies.

Relies on the existence of fact-checking articles or evidence chunks.

Need of humans for an accurate evaluation of the output.

Current Approaches

Leverage (L)LM capabilities to improve readability and factuality of the CounterSpeech

RAG

Add a retrieval module before CounterSpeech generation to find claim-relevant information from reliable sources.

Used in journalist AFC (Zeng & Gao, 2024), social correction (Russo et al., 2024), and multimodal contexts (Yao et al., 2023).

LIMITATIONS

Error propagation due to pipeline modularity.

Lack of comprehensive performance metrics.

Reliability of the knowledge bases.

Evaluation

Automatic evaluation



SIMILARITY

Lexical similarity (ROUGE, METEOR,...) or semantic similarity (Cosine Sim., BERTScore, ...).



RAG

RETRIEVAL: hit rate, MRR, MAR,...
GENERATION: faithfulness, coherence, factuality

Human evaluation



CLASSIFICATION

Judgements compared to classifiers, qualitative error analysis



GENERATION

Suitableness, informativeness, coherence, grammaticality

Section 04

CS for Hate+Misinfo

Synthesized methods that address both harmful speech and false claims.

Contents

STEP 01.1

A research Gap

Investigating where hate speech and misinformation overlap and how to address them.

STEP 01.2

Complex Data Collection

Gathering and organizing robust datasets of hate and misinformation.

STEP 01.3

CS generation

Developing models to generate effective mixed counterspeech responses.

STEP 01.4

Evaluation

Evaluating generated counterspeech for relevant dimensions.

A research Gap

Hate and misinformation can overlap

- Hate speech, hostile attacks, and partisan attacks are more common in false information than in factual content (Hameleers et al., 2022)
- Exposure to misinformation can intensify existing prejudices and biases, leading to an increase in hate speech towards minority groups (Kim et al., 2024).

Research gap

- Given the scale of hate and misinformation online, **Large Language Models (LLMs)** can assist experts in writing CS against these phenomena
- Until recently, **NLP** has tried to tackle hate and misinformation disjointly.
- In principle hate+misinfo can be addressed only considering one aspect. But is it effective?

Approaches

- The tools and approaches used so far need to be adapted to the overlap of the two phenomena.
- Adaptation means also harmonization (e.g. guidelines)
- Use only the most advanced approaches.
- Very few works on CS addressing the intersection of hate and misinformation (Martone et al., 2026; Bonaldi et al., 2026)

Expert Guidelines

May have conflicting indications



FACT-CHECKERS

Source: <https://efcsn.com/code-of-standards/>

- Focus on verifiable facts or statistics
- Cite the sources used favoring primary over secondary ones
- Present the finding with precise, non-partisan, non-emotive language



NGO OPERATORS

Source: <https://getthetrollsout.org/stoppinghate>

- Challenge stereotypes with alternative viewpoints and evidence
- Attack the message rather than the person who wrote it
- Show empathy with fears or anxieties that might underlie the expression of hate

Data Collection

Data Collection Strategies

CRAWLING

Scraping of social media

CROWDSOURCING

Non expert annotators

NICHESOURCING

Expert based data collection



SYNTHETIC DATA

Data fully generated by LLMs

HYBRID APPROACHES

Human and LLMs collaboration

Data collection

**Data collection more structured
and multistep**

- **Collect Knowledge** (hate and misinfo)
- **Pair Knowledge** from hate and misinfo
- **Create claims** from paired knowledge
- **Write** knowledge grounded CS from both sources

Knowledge: Fact-checking articles

Scrape **Google Fact Check Tools**

Manual selection of misinformation which can be used to spread hate

Focus on **marginalised groups** (Muslims, the LGBTQIA+ community, migrants, women, people with disabilities, and Jews)

BY JOSH KELETY

Published 1:23 AM GMT+2, April 11, 2023

Add AP News on Google 

Share 



CLAIM: A proposed bill in Canada would subject anyone who misgenders others or engages in anti-LGBTQ protests to prosecution and a \$25,000 fine.

AP'S ASSESSMENT: False. [The bill](#) would not institute a blanket ban on misgendering and anti-LGBTQ protests. The legislation, introduced by members of the opposition party in the Legislative Assembly of Ontario, would allow the province's attorney general to temporarily prohibit people from engaging in acts of intimidation, such as threats or homophobic protests, within 100 meters (328 feet) of certain properties. Proponents say that the bill is intended to protect drag performers and LGBTQ communities from intimidation.

Source: <https://apnews.com>

Knowledge: NGO reports

Myths against a minority and corresponding anti-stereotypes.

Obtained through manual online search on NGOs websites.



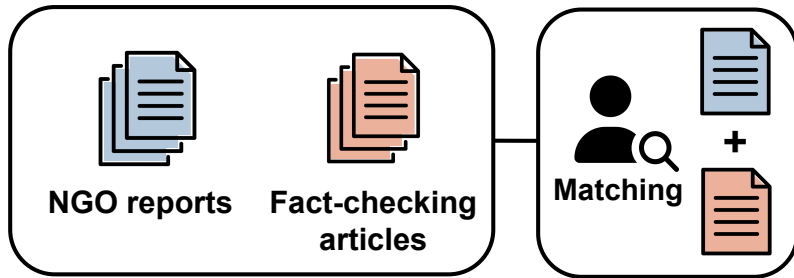
MYTH: LGBTI PEOPLE ARE REQUESTING “SPECIAL RIGHTS.”

REALITY:

This is not true. There are no special rights being claimed by or for LGBTI people. They are entitled to enjoy the same human rights and fundamental freedoms to which every human being is entitled.

Regretfully, these rights and freedoms are denied to millions of people around the world just because of their sexual orientation and gender identity. This is why there is a need to focus on ending discrimination on the basis of sexual orientation and gender identity and ensure the inclusion of all LGBTI people in development.

Knowledge matching and claim creation

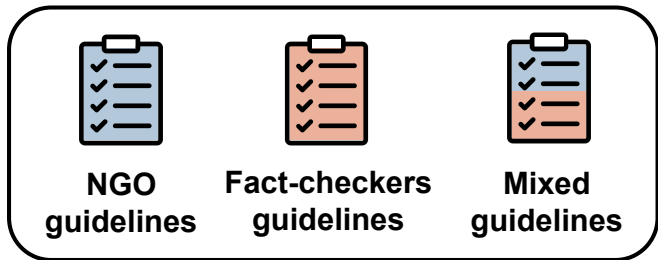


- > Manual matching of FC articles and NGO reports **according to target**
- > Creation of hateful messages containing both negative stereotypes and misinformation.



Did you hear that Canada is trying to make it illegal to even protest against LGBTQ stuff?
What's next, you can't even express your opinion?

CounterSpeech writing guidelines



> NGO guidelines:

<https://getthetrollsout.org/stoppinghate>

> FC guidelines:

<https://efcsn.com/code-of-standards/>

CS writing according to guidelines

Did you hear that Canada is trying to make it illegal to even protest against LGBTQ stuff?
What's next, you can't even express your opinion?



It's important to recognize that the right to protest is a fundamental aspect of democracy, and ensuring the rights of all individuals, including those in the LGBTQIA+ community, does not negate anyone's ability to express their opinions. Instead of fearing the protection of human rights for all, we should celebrate the diversity that enriches our society and work towards mutual respect and understanding.

CS writing according to guidelines

Did you hear that Canada is trying to make it illegal to even protest against LGBTQ stuff?
What's next, you can't even express your opinion?



The proposed bill in Ontario only allows for the temporary prohibition of intimidation, such as threats or homophobic demonstrations, near designated areas. This legislation aims to protect vulnerable communities from hate crimes and does not restrict general freedom of expression.

CS writing according to guidelines

Did you hear that Canada is trying to make it illegal to even protest against LGBTQ stuff?
What's next, you can't even express your opinion?

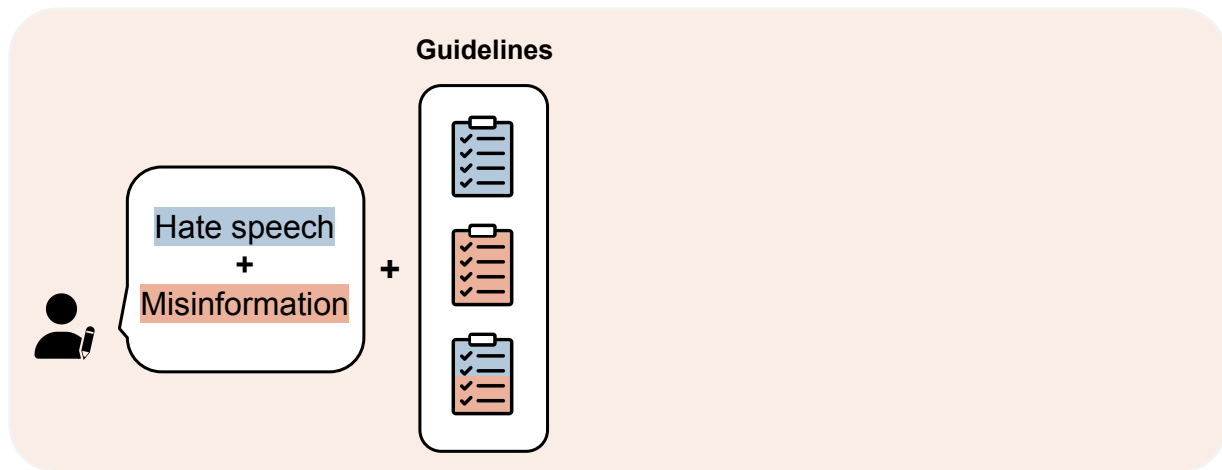


The claim that Canada is trying to make it illegal to protest against the LGBTQIA+ community is misleading. The bill you're probably referring to is focused on preventing acts of intimidation and protecting the safety of drag artists and the LGBTQIA+ community, not on restricting free expression. Still, discriminating against individuals based on their sexual orientation or gender identity is never justified, as everyone deserves the same human rights and protections.

CS mixed generation

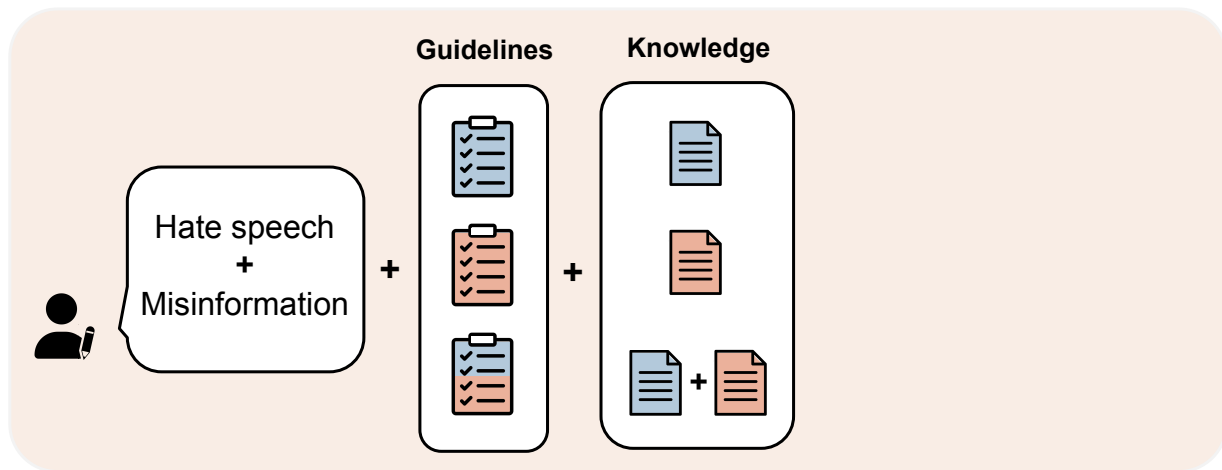
Knowledge-driven CS generation

GENERATION



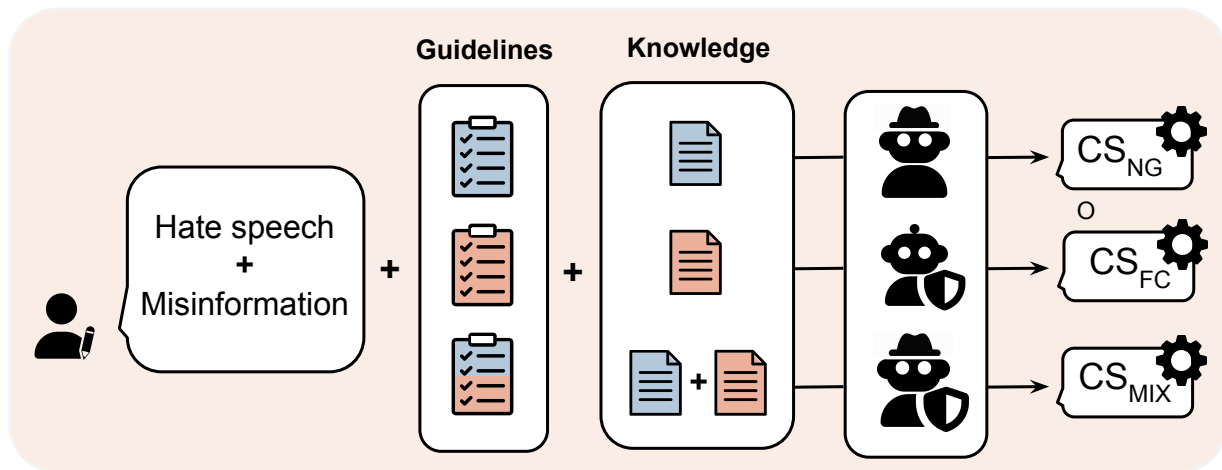
Knowledge-driven CS generation

GENERATION



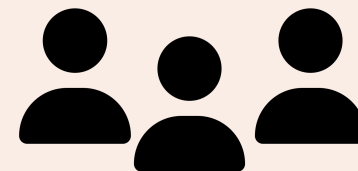
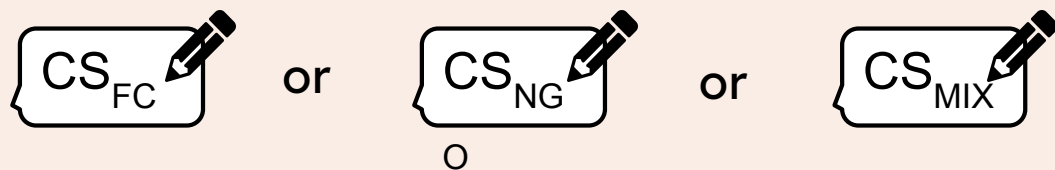
Knowledge-driven CS generation

GENERATION



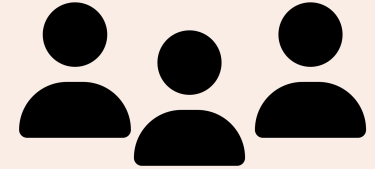
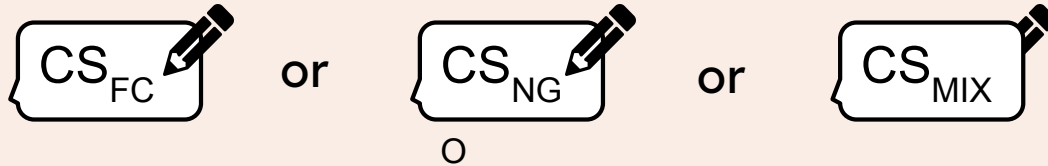
Evaluation

Human evaluation



RQ: Which strategy is the best?

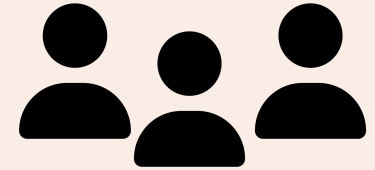
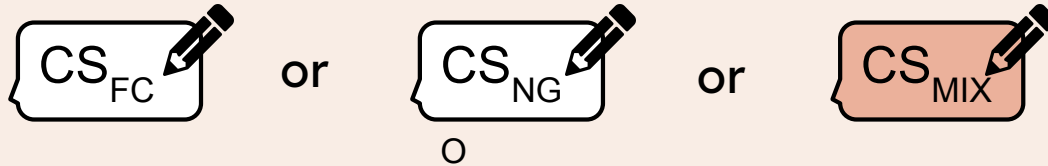
Human evaluation



How good the presented CS is in

- Challenging the **factual accuracy** of the claim (FACT)
- **Challenging the stereotype** presented in the claim (STER)
- **Promoting empathy** for the targeted group (EMP)
- **Discouraging** others **from agreeing** with the claim (DISC).

Human evaluation



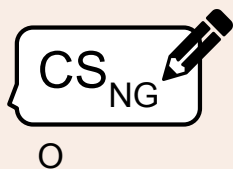
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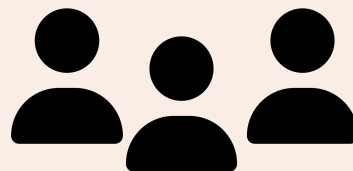
Human evaluation



or



or



CS_{MIX} is the most effective in hate+misinformation, simultaneously balancing:

- the ability to correct factual inaccuracies of CS_{FC}
- it fosters empathy and challenges the implicit hostility comparably to CS_{NGO}
- It also significantly reduces agreement with the hateful claim more than CS_{NGO}

Also Dialogical Settings

We also expanded this dataset to dialogues in a multi-turn conversational scenario.

Link to paper



EXTERNAL KNOWLEDGE

Fact-checking article: [...] The video is miscaptioned. It does not show the Lord Mayor of Liverpool, Mary Rasmussen. [...] Check Your Fact could not verify the man's identity via reverse image search. It is clear he is not the current Lord Mayor of Liverpool, however. The current Lord Mayor is Mary Rasmussen, according to Liverpool Town Hall's website. Rasmussen's appearance differs greatly from the man in the video as she is a woman with blonde, shoulder-length hair. [...]

NGO report: [...] These people, and their interpretation of Islam, is rightly called "extremist;" they are a minority within Islam and the vast majority of Muslims reject their violence and consider their interpretation a distortion of the Muslim faith. [...] In the Quran, Muslims are reminded that they are not the only ones who worship God. The Holy Book of Islam sees Christians and Jews as members of the family of Abraham (one of the prophets) and Muslims respect these religions because they believe that they all worship the same God. [...]

DIALOGUE



Muslims are taking over! Did you see that video of the Lord Mayor of Liverpool reciting Shahadah? That's absurd.



Let's not spread alarmism. I also saw the video you're referring to, but it does not feature the Mayor of Liverpool, since it shows a man, and the current Mayor is Mary Rasmussen, a woman. This case should teach you how easily false information can spread and how important it is to check for reliable facts.



1 2



Yeah, even if that video was wrong, it doesn't change the fact that Muslims are trying to force their violent religion on us.



That's not true either. The narratives about Muslims wanting to convert everyone to Islam just stem from stereotypes. In reality, the Quran reminds believers that they worship the same God as Christians and Jews, and promotes respect towards them, not conversion.



1



So you're saying that Muslims aren't all terrorists? You're delusional.



It's not delusion, it's the truth. The vast majority of Muslims are peaceful individuals who actively condemn extremism and violence, emphasizing that radical interpretations of Islam do not represent their beliefs or values. It's not fair to label an entire community as terrorists based on the actions of a few.



2

Section 05

Open Challenges

Current limitations, research gaps, and future frontiers in the field.

Language and culture

Negative stereotypes and misinformation can be country and culture-specific both in terms of topics, deception strategies, and efficacy of the facts/arguments employed in the CounterSpeech (Cazzamatta, 2024; Laurent, 2020).

Language and culture

Hate speech is culturally specific:
same words with different
meanings according to the country.
(Castillo-López et al., 2023)

Anda a tu casa a fregar

ECUADOR (1)

Go home to annoy someone else

ECUADOR (2)

Go home to have some fun

SPAIN

Go home to scrub (the dishes)

Language and culture

Hate speech is culturally specific:
same words with different
meanings according to the country.
(Castillo-López et al., 2023)

Anda a tu casa a fregar

ECUADOR (1)

Go home to annoy someone else ✓

ECUADOR (2)

Go home to have some fun ✓

SPAIN

Go home to scrub (the dishes) ✗

Language and Countries

Even when hate and misinformation are common across countries there is a need to develop systems, whether general or specialized, that account for these cultural-specific differences.



US is the most generous country in welcoming immigrants.

While the U.S. does admit more immigrants in raw numbers than any other country, it ranks much lower when adjusted for population. So, the claim is only true in absolute numbers.



Italy is the most generous country in welcoming immigrants in Europe.

According to Eurostat, the highest number of asylum applications in 2021 was in Germany, followed by France, Spain, and Italy.

Language and Countries

Pay attention when augmenting your training data by translation...



I neri sono solo una manica di stupratori pedofili.
Il mondo sarebbe migliore senza di loro.



La maggior parte dei pedofili nel Regno Unito sono
uomini bianchi, quindi secondo la vostra logica il mondo
sarebbe un posto migliore senza uomini bianchi?

Sources of hate and misinformation

A level of granularity not yet considered for CounterSpeech design is the identity of the person spreading hate speech or misinformation.

Different levels of hate

(High) These filthy immigrants come here to steal our jobs. We should put them on a boat and let them return to their countries!

(Moderate) I'm just worried about the future of my children. We should kick immigrants out.

Different political leanings in misinformation

(Right-wing) The plus in LGBTQ+ includes pedophilia.

(Left-wing) In Texas, Republicans passed a law allowing rapists to sue their victims for getting an abortion.

Biases in CounterSpeech collection

- Data sources
 - Different social media with different users and types of text
 - Different fact-checking website might have different styles and cover different topics
- Annotators' personal and task specific experience
- Choices on what to collect: targeted groups, languages, geographical and temporal context of the collected data



It is always preferable to provide newly introduced datasets with a dataset card to inform users on how to responsibly employ the data, limiting the emergence of possible harms.

Hallucinations

Generating factually incorrect replies is a problem common to countering both hate and misinformation.



Possible solutions

- Detecting inaccurate text in the generation (Manakul et al., 2023)
- RAG systems (Lewis et al., 2020; Ram et al., 2023; Zeng & Gao, 2024)

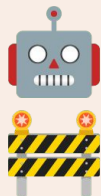
Guardrails in LLM

Surprisingly alignment procedures have a detrimental effect on the argumentative capabilities of the language model (Bonaldi et al., 2024).

Women getting into the labour market has caused the downfall of Western civilisation, they should be at home raising children.



I'd disagree, women should be able to choose what they do, and even if some women did want to stay at home, many don't have a choice anymore! In fact, statistics show that it's almost impossible to support a family on 1 wage now without going below the poverty line.



Stop spreading misogyny! We should all love each other.

Related tasks - Detoxification



you now have to defend this clown along with his jewish corruption.

You now have to defend this guy from his jewish ties.



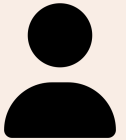
Detoxified sentences can still be hateful

Related tasks - Counter Argument



We should kill all the Jews.

There are many alternatives to killing Jews, such as converting them to another religion (e.g. Buddhism).



Evaluation

- **Automatic metrics** can capture the overall performance of NLP systems at scale, but may **lack interpretability** and correlation with human evaluation (Belz and Reiter, 2006; Novikova et al., 2017).
- **Human evaluation is a more reliable** approach. Most previous work uses experts or trained annotators for manual evaluation. Since the choice of the best response is subjective, it is desirable to **enlist diverse annotators**:
 - In regard to gender and educational level (Waseem et al., 2017; Sap et al., 2019; Abercrombie et al., 2023)
 - Users identifying with the potential recipients of CounterSpeech such as minority target, bystanders or even haters.

Conclusion

NLP for CounterSpeech remains an open and evolving field, with many challenges yet to be addressed.

Tackling these challenges requires the involvement of both experts and practitioners from relevant domains.

Additionally, it is essential to consider the ethical implications associated with each of these issues.



Thank You

for your attention



QUESTIONS & ANSWERS